The extracted wave file shows the value of Ascalon Cluster level I/O Interfaces used for reset. Signals of interest are:

Clocks
- i_fb_clk
- i_rf_clk
- i_sc_clk

Reset/Interrupts/Control Signals
- i_cluster_cold_reset_n
- i_force_ss_to_ref_clock_n
- o_cluster_interrupts[18] (config done interrupt)

CPL AXI4 Interface
- **Write Address Channel**
  - i_cpl_s0_awvalid
  - o_cpl_s0_awready
  - i_cpl_s0_awid[6:0]
  - i_cpl_s0_awaddr[31:0]
  - i_cpl_s0_awlen[7:0]
  - i_cpl_s0_awsize[2:0]
  - i_cpl_s0_awburst[1:0]
  - i_cpl_s0_awprot[2:0]
  - i_cpl_s0_awuser[3:0]
- **Write Data Channel**
  - i_cpl_s0_wvalid
  - o_cpl_s0_wready
  - i_cpl_s0_wdata[63:0]
  - i_cpl_s0_wstrb[7:0]
  - i_cpl_s0_wlast
- **Write Response Channel**
  - i_cpl_s0_bready
  - o_cpl_s0_bid[6:0]
  - o_cpl_s0_bresp[1:0]
  - o_cpl_s0_bvalid

The attached “rc” file shows different markers for reset and initialization steps.

Time (in fs)	Marker Name	Description
315200000	“Out of Cold Reset”	SMC deasserts cluster_cold_reset_n
535200000	“Remove force_ss_to_ref_clk”	SMC set force_ss_to _ref_clock_n to ‘0’to run all clocks with REF clock
701600000	“SRAM Wr CrClusterFuse”	SMC programs Fuse config
738800000	“SRAM Wr Reset Vector”	SMC configures CPL reset vector
776400000	“Reset Ctrl Wr (Release CPL out of Reset)”	SMC programs CPL reset ctrl to start CPL Firmware execution
71866250000	“Config Done”	CPL sends Config Done Interrupt to SMC
71933750000	“Write reg to program ACLINT mtime”	CPL updates System timer
72098750000	“Write reg to allow Core fetch”	SMC sends start execution of core to CPL